

Spring 2009

Problem 1 (Sp09, Sp18). Suppose that f is a continuous real function on $[0, 1]$. Prove that

$$\lim_{\alpha \rightarrow 0^+} \alpha \int_0^1 x^{\alpha-1} f(x) dx = f(0)$$

Problem 2 (Sp09). Prove that if an $n \times n$ matrix X over $\mathbb{R}$ satisfies $X^2 = -I$, then n is even.

Problem 3 (Sp09, Sp15). Show that if $f : \mathbb{C} \rightarrow \mathbb{C} \cup \infty$ is a meromorphic function in the plane, such that there exist $R, C > 0$ so that for $|z| > R$, $|f(z)| \leq C|z|^n$, then f is a rational function.

Problem 4 (Sp09). Let G be a finite group, and for each positive integer n , let

$$X_n = \{(g_1, \dots, g_n) \mid g_i g_j = g_j g_i \text{ for all } i, j\}$$

Show that the formula

$$h \cdot (g_1, \dots, g_n) = (hg_1h^{-1}, \dots, hg_nh^{-1})$$

defines an action of G on X_n , and that $|X_{n+1}| = |G| \cdot |X_n/G|$ for all n , where X_n/G denotes the set of G -orbits in X_n .

Problem 5 (Sp09). There is a *folk theorem* that a four-footed table can always be rotated into a stable position on an uneven floor. Prove the following mathematical formulation of this theorem:

Define four points in $\mathbb{R}^2$, depending on an angle θ , by $P_1(\theta) = (\cos \theta, \sin \theta)$, $P_2(\theta) = (-\sin \theta, \cos \theta)$, $P_3(\theta) = (-\cos \theta, -\sin \theta)$, $P_4(\theta) = (\sin \theta, -\cos \theta)$. Show that given any continuous function $h: \mathbb{R}^2 \rightarrow \mathbb{R}$, there exists a value of θ such that the four points $Q_i(\theta) = (P_i(\theta), h(P_i(\theta)))$ on the graph of h are coplanar in $\mathbb{R}^3$.

Problem 6 (Sp09). Let $M_2(\mathbb{C})$ be the set of 2×2 matrices over the complex numbers. Given $A \in M_2(\mathbb{C})$, define $C(A)$ be the *commutant* of A as $C(A) = \{B \in M_2(\mathbb{C}) \mid AB = BA\}$.

1. Prove that $C(A)$ is a linear subspace of $M_2(\mathbb{C})$, for every A .
2. Determine all possible values of the dimension $\dim C(A)$.

3. Formulate a simple and explicit rule to find $\dim C(A)$, given A .

Problem 7 (Su77, Fa84, Sp94, Sp96, Sp09, Fa11). Evaluate the integral

$$I(a) = \int_0^{2\pi} \frac{d\theta}{a + \cos \theta}$$

where a is real and $a > 1$. Why the formula obtained for $I(a)$ is also valid for certain complex (nonreal) values of a ?

Problem 8 (Sp05, Sp09). Let $\mathbb{Q}(x)$ be the field of rational functions of one variable over $\mathbb{Q}$. Let $i: \mathbb{Q}(x) \rightarrow \mathbb{Q}(x)$ be the unique field automorphism such that $i(x) = x^{-1}$. Prove that the subfield of elements fixed by i is equal to $\mathbb{Q}(x + x^{-1})$.

Problem 9 (Sp09). Let $d_k = \text{lcm}\{1, 2, \dots, k\}$ and

$$I_m = \int_0^1 x^m (1-x)^m dx$$

Show that $d_{2m+1} I_m$ is an integer, and use this to show that $d_{2m+1} \geq 2^{2m}$.

Problem 10 (Sp05, Sp09). Let I be an open interval and let $f: I \rightarrow \mathbb{R}$ have continuous k -th derivatives everywhere on I for all $k \leq n-1$. Let $a \in I$ be such that $f^{(k)}(a) = 0$ for $1 \leq k \leq n-1$, and assume that $f^{(n)}(a)$ is defined and $f^{(n)}(a) > 0$. Prove that if n is even, then f has a local minimum at a , and if n is odd, then f has no local extremum at a .

Problem 11 (Sp09). Let A and B be $n \times n$ matrices over a field of characteristic zero. Prove that the condition $BA - AB = A$ implies that A is nilpotent.

Problem 12 (Fa92, Sp09). 1. How many zeros does the polynomial defined by $p(z) = 2z^5 + 4z^2 + 1$ have in the disc $|z| < 1$?

2. How many zeros does the same polynomial have on the real axis?

Problem 13 (Sp09). Consider a polynomial expression $H(\alpha) = q_0 + q_1\alpha + q_2\alpha^2 + \dots + q_n\alpha^n$ with rational coefficients $q_0, q_1, q_2, \dots, q_n$, where α is an algebraic number. Prove that if $H(\alpha) \neq 0$, then the reciprocal $1/H(\alpha)$ can be expressed as a polynomial in α with rational coefficients.

Problem 14 (Sp09, Fa04). Let $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ be a continuous function of compact support. Show that

$$u(x) = \int_{\mathbb{R}^3} \frac{f(y)}{\|x - y\|} dy$$

is well defined and that

$$\lim_{\|x\| \rightarrow \infty} u(x)\|x\| = \int_{\mathbb{R}^3} f(y) dy$$

Problem 15 (Sp09). Let $\lambda_1 \geq \dots \geq \lambda_n$ be eigenvalues of a symmetric real $n \times n$ matrix A . Prove that

$$\lambda_k = \max_{V^k} \min_{x \in V^k \setminus \{0\}} \frac{\langle Ax, x \rangle}{\langle x, x \rangle}$$

where the maximum is taken over all k -dimensional linear subspaces V^k , the minimum over all nonzero vectors in the subspace.

Problem 16 (Sp09). If f is a univalent (1-1 analytic) function with domain the unit disc such that $f(z) = z + \sum_{n=2}^{\infty} a_n z^n$, then prove that

$$g(z) = \sqrt{f(z^2)}$$

is an odd analytic univalent function on the unit disc.

Problem 17 (Fa97, Sp09, Sp16, Sp26). Let G be a finite group.

1. If G is non-abelian, show that $G/Z(G)$ is not cyclic, where $Z(G)$ is the center of G .
2. If $|G| = p^n$, with p prime and $n > 0$, show that $Z(G)$ is not trivial and deduce that G has a normal subgroup of order p .
3. If $|G| = p^2$, show that G is abelian.

Problem 18 (Sp09). Prove that the sequence of functions $f_n(x) = \sin nx$ has no pointwise convergent subsequence.